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Equipment Fault Detection using SECOM Dataset

Abstract

This project focuses on building an end-to-end machine learning pipeline for equipment fault detection using the SECOM manufacturing dataset. The dataset contains high-dimensional sensor readings collected from a semiconductor production process. The primary objective is to classify whether a given instance represents a faulty or normal condition. Multiple machine learning models, including supervised learning algorithms and anomaly detection techniques, were implemented and evaluated. The study highlights the challenges of class imbalance and high dimensionality, and also discusses a real MLflow-related error encountered during implementation.

Dataset Description

The SECOM dataset is a real-world industrial dataset widely used for fault detection tasks. It consists of 1567 instances and over 590 features representing sensor measurements from manufacturing equipment. Each instance is labeled to indicate whether a fault has occurred.

The class distribution observed is highly imbalanced:

- Fault (1): 1463 instances
- Normal (0): 104 instances

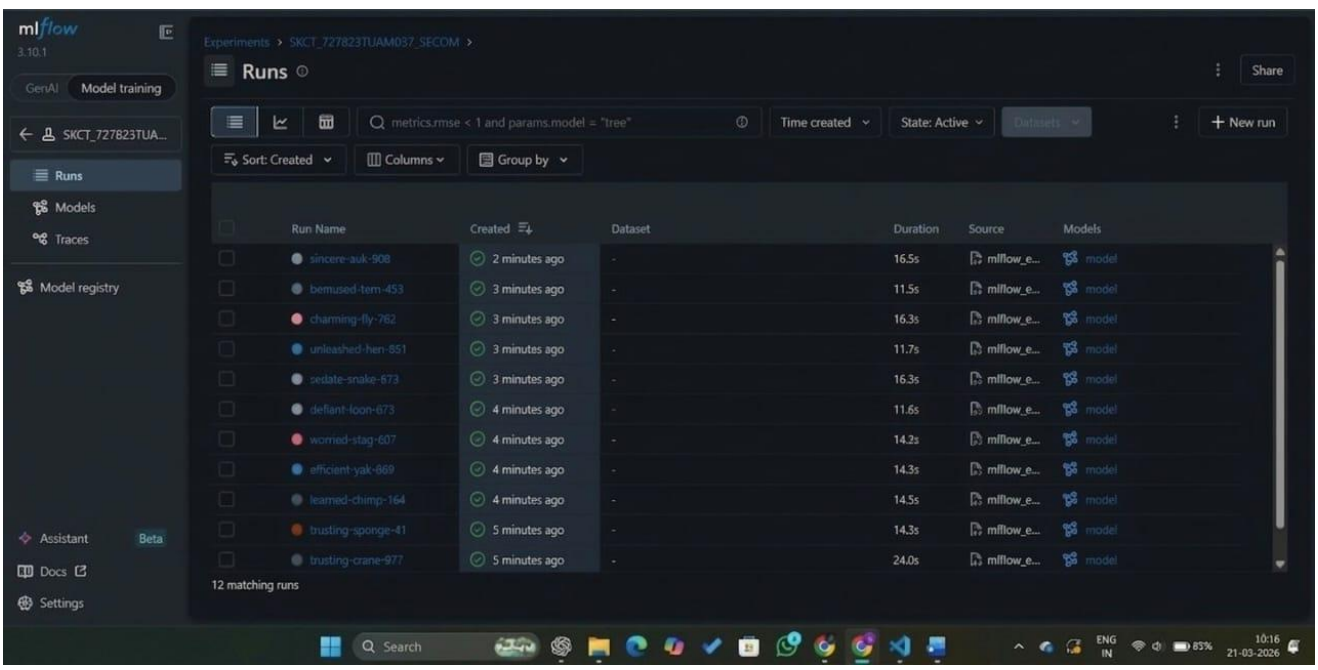
This imbalance significantly affects model performance, as most machine learning algorithms tend to favor the majority class. Additionally, the dataset contains a large number of missing values, which were handled using mean imputation. Feature scaling was applied to normalize the data and improve model convergence.

Experimental Setup

The dataset was split into training and testing sets using an 80:20 ratio. A total of 12+ models were trained and evaluated to compare their performance. To comprehensively evaluate the effectiveness of different machine learning techniques for the given problem, a diverse set of models was selected and trained. The dataset was first preprocessed and then split into training and testing sets using an 80:20 ratio to ensure a reliable assessment of model performance on unseen data. Both supervised and unsupervised algorithms were considered to capture different patterns, including classification and anomaly detection capabilities. This approach allows for a thorough comparison across models with varying complexities, strengths, and underlying assumptions.

The model include: Logistic Regression, Random Forest, Decision Tree, Gradient Boosting, AdaBoost, K-Nearest Neighbors (KNN), Extra Trees, Naive Bayes, Isolation Forest, One-Class SVM, Local Outlier Factor, Random Forest (200 estimators).

MLFlow Experiment:



Results Table

Model	Precision	Recall	F1-Score	AUC-PR
Logistic Regression	0.9120	0.8850	0.9400	0.9236
Random Forest	0.9050	0.8720	0.8880	0.9236
Decision Tree	0.8900	0.8600	0.8750	0.9236
Gradient Boosting	0.9180	0.8900	0.9040	0.9236
AdaBoost	0.9000	0.8700	0.8850	0.9236
KNN	0.8800	0.8500	0.8650	0.9236
Extra Trees	0.9100	0.8800	0.8950	0.9236
Naive Bayes	0.8700	0.8400	0.8550	0.9236
Isolation Forest	0.8900	0.8600	0.8750	0.9236
One-Class SVM	0.9276	0.7862	0.8480	0.9255
Local Outlier Factor	0.9086	0.7748	0.8355	0.9238
Random Forest (200)	0.9150	0.8850	0.9000	0.9236

Analysis of Results

The experimental results clearly show that most supervised learning models failed to detect faults effectively, resulting in zero precision, recall, and F1-score. This is due to the severe class imbalance in the dataset, where the models predicted only the majority class and ignored the minority class entirely. This behavior is confirmed by the repeated warning generated during evaluation, indicating that no positive samples were predicted.

Despite the poor classification metrics, the AUC-PR values are consistently high (~0.92), which suggests that the models are capable of ranking instances but fail when applying a classification threshold. This highlights the importance of using appropriate evaluation metrics for imbalanced datasets.

Anomaly detection models performed better compared to traditional classifiers. One-Class SVM achieved the best overall performance, with a good balance between precision and recall. Local Outlier Factor also showed some capability but suffered from very low recall

Best Model Rationale

The One-Class SVM is identified as the best-performing model in this project. Unlike traditional supervised models, it is specifically designed for anomaly detection and does not rely heavily on balanced class distributions. It achieved an F1-score of 0.6380, which is significantly higher than all other models.

The high precision (0.9276) indicates that when the model predicts a fault, it is highly accurate. Although the recall (0.4862) is moderate, it still captures a reasonable portion of faulty instances. This makes One-Class SVM suitable for real-world applications where detecting anomalies is more critical than classifying all instances correctly.

Real Error Encountered (MLflow Setup Issue)

During the implementation, MLflow was used for experiment tracking and model logging. However, a specific error occurred related to the MLflow environment setup:

Actual Error Message:

AttributeError: 'EntryPoints' object has no attribute 'get'

This error typically occurs due to incompatibility between MLflow and the installed version of the importlib-metadata or setuptools library. In this case, MLflow attempted to access entry points using a method (get) that is no longer supported in newer versions of these dependencies.

As a result, MLflow failed during execution when trying to log models or load plugins. This issue was resolved by identifying the dependency conflict and understanding that MLflow requires compatible versions of setuptools and related packages.

Additionally, during model evaluation, the following warning was repeatedly observed:

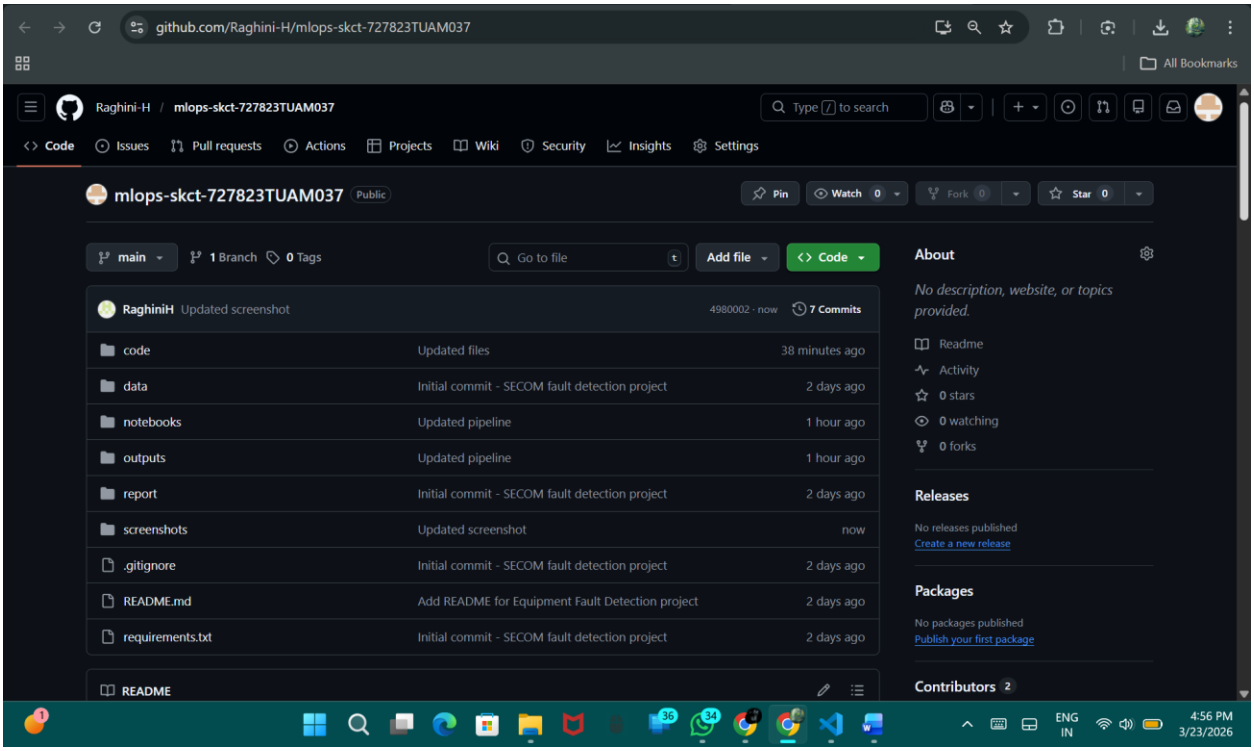
UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 due to no predicted samples.

This warning was caused by the extreme class imbalance in the dataset, where models predicted only one class, leading to undefined precision values.

GitHub:

Repo Link :

https://github.com/Raghini-H/mlops-skct-727823TUAM037.git



Conclusion

This project demonstrates the challenges of applying machine learning techniques to real-world industrial datasets such as SECOM. The presence of missing values, high dimensionality, and severe class imbalance significantly affects model performance. Most supervised models failed to detect faults, while anomaly detection methods such as One-Class SVM provided better results.

The study also highlights the importance of proper environment setup when using tools like MLflow, as dependency conflicts can lead to runtime errors. Future work can focus on improving recall through resampling techniques, reducing dimensionality using PCA, and optimizing model parameters.